Amélioration de la résilience des modèles prédictifs par la qualité et la continuité des données

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• Fundamental Principle of WNNmulti Model

WNNmulti extends the univariate Weighted Nearest Neighbours (WNN) method to a multivariate context. Its core principle is to project future values (or generate missing ones) by identifying similar patterns in historical data.

The central idea is to improve the forecast of a given series $x_i$by using information not only from its own past, but also from the past of other related time series (the other buildings and the PV production).

• Key Operational Parameters

w: Size of the historical window (input pattern). In your context, $w$corresponds to the last seven days of data.

h : Forecasting horizon. For hourly energy forecasting, typically $h = 24$hours.

T : Current time (start of the forecast).

k : Number of nearest neighbours used for prediction.

We consider a set of d multivariate time series.

- <u>Set of Time Series</u>

$$(x_1, x_2, \dots, x_d)^T$$

- <u>Time Series i</u>

Each series x_i is observed up to the current time T:

$$x_i = (x_1^i, \dots, x_T^i)$$

- <u>Forecasting Objective</u>

The goal is to forecast each of the d series over a future horizon of length h:

$$\hat{x}_i^h = \left(\hat{x}_{T+1}^i, \dots, \hat{x}_{T+h}^i\right) \quad \text{for} \quad i = 1, \dots, d$$

- ## Segmentation and Pattern Definition (Univariate WNN Basis)

The WNNmulti method builds on the univariate WNN approach by segmenting historical data

For clarity, the index iis temporarily omitted.

- <u>Target Pattern</u>

The target pattern corresponds to the last w values at time T:

$$x_{T-w+1,T} = (x_{T-w+1}, \dots, x_T)$$

- <u>Set of Past Patterns</u>

The historical pattern set is defined as :

$$E_p = \{x_{j-w+1,\,j} \;\mid\; j = w, \dots, T-h\}$$

- <u>Set of Future Sequences</u>

Each past pattern has an associated future sequence of length h:

$$E_f = \{x_{j+1,\,j+h} \;\mid\; j = w, \dots, T-h\}$$

- ## WNNmulti Algorithm (Multivariate Extension)

WNNmulti extends the WNN approach by allowing the nearest-neighbour search to span all dseries.

- <u>Multivariate Historical Sets</u>

The historical datasets include all patterns from all series:

- **All past patterns:**

$$E_p^{all} = \{x_{j-w+1,\,j}^{i'} \;\mid\; j = w, \dots, T-h \;\; and \;\; i' = 1, \dots, d\}$$

- **All future sequences:**

$$E_f^{all} = \{x_{j+1,\,j+h}^{i'} \;\mid\; j = w, \dots, T-h \;\; and \;\; i' = 1, \dots, d\}$$

- <u>Distance Computation</u>

For a target series i, the Euclidean distance between its target pattern

$$x_{T-w+1,T}^{i}$$

and each multivariate historical pattern is computed:

$$d_j = \| x^{i}_{T-w+1,T} - x^{i'}_{j-w+1,j} \|_2$$

where i' denotes the series from which the historical pattern originates.

- Selection of the **k**Nearest Neighbours

The indices corresponding to the k smallest distances form the neighbour set:

$$J_k^{all} = \{(i'_1, j_1), (i'_2, j_2), \dots, (i'_k, j_k)\}$$

Each tuple specifies the neighbour's series index i'_k and historical time index j_k.

- Weighted Forecasting

The forecast $\hat{x}_i^h$ is calculated as a weighted average of the future sequences associated with the k selected neighbours.

Weighting Scheme

Each neighbour receives a weight inversely proportional to the square of its distance:

$$\alpha_{(i',j)} = \frac{1}{(d_j)^2}$$

WNNmulti Forecast Formula

$$\hat{x}_i^h = \frac{\sum_{(i',j)\in J_k^{all}} \alpha_{(i',j)}\, x^{i'}_{j+1,\, j+h}}{\sum_{(i',j)\in J_k^{all}} \alpha_{(i',j)}}$$

This ensures that the most similar historical patterns contribute the most to the final prediction.

• Advantages for the LiveTree Project

Exploitation of Multivariate Dependencies
WNNmulti naturally integrates information from related series (building consumption and PV production). It can detect local similarities even when correlations are not globally strong.

Simplicity and Flexibility
As a non-parametric method, it avoids explicit modeling of nonlinear interactions. It also allows exploring alternative similarity measures and weighting strategies.

Computational Efficiency
WNNmulti runs significantly faster than many recursive multivariate ML approaches such as XGBoost or Random Forest and approaches the speed of the most efficient methods tested (e.g., MV-kWNN).

- ## Limitations

Potential Degradation of Accuracy
In cases of high data heterogeneity or the presence of rare events, the inclusion of "nearby" but irrelevant historical patterns can distort the distance metric and reduce accuracy (as observed during days with the highest RMSE values).

Dependence on Historical Data
The method requires a sufficiently long and representative historical dataset in order to identify meaningful past patterns resembling current conditions.

BIBLIOGRAPHIE

https://research-ebsco-com.ezproxy.univ-catholille.fr/c/z5kttp/search/details/ehyubswhjr?db=bth

https://doaj.org/article/cf40a90163134f3aad734641a64f7b6c#

https://www-sciencedirect-com.ezproxy.univ-catholille.fr/science/article/pii/S0020019009003123?via%3Dihub

https://link-springer-com.ezproxy.univ-catholille.fr/article/10.1023/A:1022664626993

https://www-jstor-org.ezproxy.univ-catholille.fr/stable/41806553?seq=17

https://pubmed.ncbi.nlm.nih.gov/24895665/

https://arxiv.org/abs/2110.13031

https://digital.library.unt.edu/ark:/67531/metadc723200/m1/?q=Final%20Report:%20Weighted%20Neighbor%20Data%20Mining

https://ieeexplore.ieee.org/abstract/document/6645422

https://ieeexplore.ieee.org/abstract/document/8006263

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